

SOLIDWORKS API (.NET VBA C#) — Reference (48/337): IComponent2 (part 1 of 4)

SOLIDWORKS API Documentation (converted from CHM)

IComponent2 (part 1 of 4) (Type)

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IComponent2 Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : IComponent2 Interface

Allows access to the components within assemblies.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

```
Public Interface IComponent2
```

Visual Basic (Usage)



```
Dim instance As IComponent2
```

C#

```
public interface IComponent2
```

C++/CLI

```
public interface class IComponent2
```

Visual Basic for Applications (VBA) Syntax

See [Component2](#).

Example

[Insert New Virtual Component \(C#\)](#)
[Insert New Virtual Component \(VB.NET\)](#)
[Insert New Virtual Component \(VBA\)](#)
[Insert MidSurface in Component \(C#\)](#)
[Insert MidSurface in Component \(VB.NET\)](#)
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[Get Model Textures \(VBA\)](#)
[Get Model Textures \(VB.NET\)](#)
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[Get Model Material Property Values \(VBA\)](#)
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[Get Model Material Property Values \(C#\)](#)
[Get Referenced Display State \(VBA\)](#)
[Get Referenced Display State \(VB.NET\)](#)
[Get Referenced Display State \(C#\)](#)

Remarks

Use this interface to traverse assemblies.

Information returned from the methods and properties of this interface is based on the current configuration and might vary if the user displays a different configuration.

Accessors

[IAssemblyDoc::AddComponents3](#) and [IAssemblyDoc::IAddComponents2](#)
[IAssemblyDoc::AddComponent5](#)
[IAssemblyDoc::AddPLMComponent](#)
[IAssemblyDoc::GetComponentById](#)
[IAssemblyDoc::GetComponents](#) and [IAssemblyDoc::IGetComponents](#)
[IAssemblyDoc::GetComponentByName](#)
[IAssemblyDoc::GetEditTargetComponent](#)
[IAssemblyDoc::GetFeatureScope](#) and [IAssemblyDoc::IGetFeatureScope](#)

[IAssemblyDoc::GetVisibleComponentsInView](#) and [IAssemblyDoc::IGetVisibleComponentsInView](#)
[IAssemblyDoc::InsertEnvelope](#)
[IAssemblyDoc::MirrorComponents2](#)
[IAssemblyDoc::ToolsCheckInterference2](#) and [IAssemblyDoc::IToolsCheckInterference3](#)
[IAttribute::GetComponent](#) and [IAttribute::IGetComponent2](#)
[IBomTableAnnotation::GetComponents2](#) and [IBomTableAnnotation::IGetComponents2](#)
[ICavityFeatureData::Components](#)
[ICavityFeatureData::IGetComponents](#)
[IChainPatternFeatureData::Group1PatternComponent](#)
[IChainPatternFeatureData::Group2PatternComponent](#)
[IClearanceResult::ComponentsOrFaces](#)
[IClearanceVerificationMgr::GetComponentsOrFacesToCheck](#)
[ICollision::GetComponents](#)
[ICollisionDetectionGroup::GetComponents](#)
[IComponent2::GetChildren](#) and [IComponent2::IGetChildren](#)
[IComponent2::GetParent](#)
[IConfiguration::GetDisplayStateComponentProperties](#)
[IConfiguration::GetDisplayStateComponentVisibility](#)
[IConfiguration::GetRootComponent3](#)
[IDrawingComponent::Component](#)
[IEntity::GetComponent](#) and [IEntity::IGetComponent2](#)
[IEnumComponents2::Next](#)
[IExplodeStep::GetComponent](#) and [IExplodeStep::IGetComponent](#)
[IExplodeStep::GetComponents](#)
[IHoleSeriesFeatureData2::GetComponents](#) and [IHoleSeriesFeatureData2::IGetComponents](#)
[IInterference::Components](#) and [IInterference::IGetComponents](#)
[IInterferenceDetectionMgr::GetComponentsAndTransforms](#)
[IInterferenceDetectionMgr::GetInterferenceComponents](#) and [IInterferenceDetectionMgr::IGetInterferenceComponents](#)
[IJoinFeatureData::IGetJoinedParts](#)
[IJoinFeatureData::JoinedParts](#)
[ILinearCouplerMateFeatureData::ReferenceComponent1](#)
[ILinearCouplerMateFeatureData::ReferenceComponent2](#)
[ILocalCurvePatternFeatureData::PatternComponentArray](#)
[ILocalSketchPatternFeatureData::PatternComponentArray](#)
[IMassProperty2::SelectedItems](#)
[IMateEntity2::ReferenceComponent](#)
[IMateInPlace::Component](#)
[IMateLoadReference::Component](#)
[IMirrorComponentFeatureData::ComponentsToInstanceAlignToComponentOrigin](#)
[IMirrorComponentFeatureData::OppositeHandComponents](#)
[ISectionViewData::SelectiveSectioningList](#)
[ISectionViewData::TransparencyList](#)
[ISelectionMgr::GetSelectedObjectsComponent3](#)
[ISimulationSpringFeatureData::BaseComponent](#)
[IView::GetVisibleComponents](#) and [IView::IGetVisibleComponents](#)

Access Diagram

[Component2](#)

See Also

[IComponent2 Members](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IModelDoc2::GetPathName Method \(\)](#)

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IComponent2 Interface Members

[See Also](#) [Properties](#) [Methods](#)

[SolidWorks.Interop.sldworks Namespace](#) : IComponent2 Interface

The following tables list the members exposed by [IComponent2](#).

Public Properties

Name	Description
 ComponentReference	Gets or sets a component reference for this component.
 CustomPropertyManager	Gets the configuration-specific custom property manager for this assembly component.
 DisplayTitle	Gets this component's title as displayed in the FeatureManager design tree.
 ExcludeFromBOM	Obsolete. Superseded by IComponent2::GetExcludeFromBOM2 and IComponent2::SetExcludeFromBOM2 .
 IMaterialPropertyValues	Gets or sets the material properties for the selected component in the active configuration.
 IsGraphicsOnly	Gets whether this component is graphics only.
 IsSpeedPak	Gets whether the active configuration for this component is SpeedPak.
 IsVirtual	Gets whether this component is a virtual component. NOTE: This property is a get-only property. Set is not implemented.
 MaterialPropertyValues	Gets or sets the material properties for the selected component in the active configuration.
 Name	Obsolete. Superseded by IComponent2::Name2 .
 Name2	Gets or sets the name of the selected component.
 PresentationTransform	Gets or sets the component transform.
 ReferencedConfiguration	Gets or sets the active configuration used by this component.
 ReferencedDisplayState	Obsolete. Superseded by IComponent2::ReferencedDisplayState2 .
 ReferencedDisplayState2	Gets or sets the active display state of this component.
 Solving	Gets the Solve as option (rigid or flexible) of this component.
 Transform	Obsolete. Superseded by IComponent2::Transform2 .
 Transform2	Gets or sets the component transform.
 UseNamedConfiguration	Gets whether a specified configuration or the in-use/last active configuration is used.
 Visible	Gets or sets the visibility state of this component.

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Public Methods

Name	Description
 AddPropertyExtension	Adds a property extension to this component.
 DeSelect	Deselects this component.
 EnumBodies	Obsolete. Superseded by IComponent2::EnumBodies2 .
 EnumBodies2	Obsolete. Superseded by IComponent2::EnumBodies3 .
 EnumBodies3	Gets the bodies in the component in a multibody part.
 EnumRelatedBodies	Creates an enumerated list of bodies.
 EnumSectionedBodies	Gets the sectioned bodies seen in the specified view and returns them in an enumerated list.
 FeatureByName	Gets the specified feature for this component.

		
	FindAttribute	Finds an attribute on a component.
	FirstFeature	Gets the first feature in this component.
	GetBodies	Obsolete. Superseded by IComponent2::GetBodies2 .
	GetBodies2	Obsolete. Superseded by IComponent2::GetBodies3 .
	GetBodies3	Gets the bodies in this component.
	GetBody	Gets the body that belongs to this instance of this component.
	GetBox	Gets the bounding box for component.
	GetChildren	Gets all of the children components of this component.
	GetConfigurationCount	Gets the number of configurations for this lightweight assembly component.
	GetConfigurationNames	Gets the names of configurations for this lightweight assembly component.
	GetConstrainedStatus	Gets the constrained status of this component.
	GetCorresponding	Gets the corresponding object in the context of the assembly for a specific instance of the component.
	GetCorrespondingEntity	Gets the entity that corresponds with the Dispatch pointer in the context of the component.
	GetDecals	Gets the decals applied to this component.
	GetDecalsCount	Gets the number of decals applied to this component.
	GetDrawingComponent	Gets the drawing component for this component.
	GetExcludeFromBOM2	Gets whether this component is excluded from the bills of materials (BOMs) in the specified configurations.
	GetHiddenUnloadedChildrenCount	Gets the number of hidden children components of this component that were not loaded when an assembly was opened selectively .
	GetID	Gets the component ID for this component.
	GetImportedPath	Gets the full path name of the model imported for this component.
	GetMaterialIdName	Gets the material name for this component.
	GetMaterialPropertyValues2	Gets the material properties for this component.
	GetMaterialUserName	Gets the user-visible name of the material for this component.
	GetMates	Gets the mates for this component.
	GetModelDoc	Obsolete. Superseded by IComponent2::GetModelDoc2 .
	GetModelDoc2	Gets the model document for this component.
	GetModelMaterialPropertyValues	Gets the material properties of this lightweight component in the specified configuration.
	GetModelTexture	Gets the texture applied to this lightweight component in the specified configuration.
	GetParent	Gets the parent component.
	GetPathName	Gets the full path name for this component.
	GetPLMID	Gets the ID of this SOLIDWORKS Design connected to the 3DEXPERIENCE platform document.
	GetPropertyExtension	Gets the property extension on this component.
	GetReferencedDisplayStates	Gets the display states of this component that are referenced by the specified assembly display state(s).
	GetRenderMaterials	Obsolete. Superseded by IComponent2::GetRenderMaterials2 .
	GetRenderMaterials2	Gets the appearances applied to this component in the specified display states.
	GetRenderMaterialsCount	Obsolete. Superseded by IComponent2::GetRenderMaterialsCount2 .
	GetRenderMaterialsCount2	Gets the number of appearances applied to this component in the specified display states.
	GetRepresentationParent	Gets the representation parent of this SOLIDWORKS Design connected to the 3DEXPERIENCE platform document.
	GetSectionedBodies	Gets the sectioned bodies in the specified section view.
	GetSelectByIdString	Gets the name of the component for possible use with IModelDocExtension::SelectById2 , for selectively opening a document using ISldWorks::OpenDoc7 and IDocumentSpecification , etc.
	GetSmartComponentData	Gets the features, components, and feature references of a Smart Component.
	GetSpecificTransform	Get the collapsed or exploded transform of a component when the assembly is exploded.

		
	GetSuppression	Obsolete. Superseded by IComponent2::GetSuppression2
	GetSuppression2	Gets the suppression state of this component.
	GetTessNorms	Gets the normal vector for each of the triangles, which make up the shaded picture tessellation for the component.
	GetTessTriangles	Gets the triangles that make up the shaded picture tessellation for this component.
	GetTessTriStripEdges	Gets the edge IDs for the triangle strips.
	GetTessTriStripNorms	Gets the normal vector for each of the triangles, which make up the shaded picture tessellation for this component.
	GetTessTriStrips	Gets the vertices that make up the shaded picture tessellation for this component.
	GetTexture	Gets the texture applied to this component in the specified configuration.
	GetTotalTransform	Combines the original transform of this component with the presentation transform of this component.
	GetType	Gets this lightweight assembly component's document type.
	GetUnloadedComponentNames	Gets the component's unloaded children components' path names, referenced configuration names, reasons why they are unloaded, document types, and names.
	GetVisibility	Gets the visibility state for this component.
	GetVisibilityInAsmDisplayStates	Gets the visibilities of this component in the specified assembly display state(s).
	GetXform	Obsolete. Superseded by IComponent2::Transform2
	HasAssemblyFeatures	Gets whether this component has assembly features in the specified configuration.
	HasMaterialPropertyValues	Gets whether this component has an appearance.
	HasUnloadedComponents	Gets whether this component has hidden or suppressed unloaded children components.
	IFindAttribute	Finds an attribute on a component.
	IGetBody	Gets the body that belongs to this instance of this component.
	IGetBox	Gets the bounding box for component.
	IGetChildren	Gets all of the children components of this component.
	IGetChildrenCount	Gets the number of children components for this component.
	IGetCorrespondingEntity	Gets the entity that corresponds with the Dispatch pointer in the context of the component.
	IGetDecals	Gets the decals applied to this component.
	IGetMaterialPropertyValues2	Gets the material properties for this component.
	IGetMaterialPropertyValuesForFace	Gets the color of the specified face.
	IGetModelDoc	Obsolete. Superseded by IComponent2::GetModelDoc2 .
	IGetModelMaterialPropertyValues	Gets the material properties of this lightweight component in the specified configuration.
	IGetRenderMaterials	Obsolete. Superseded by IComponent2::GetRenderMaterials2 .
	IGetTemporaryBodyID	Gets the current body tag ID, which is not a permanent ID.
	IGetTessNorms	Gets the normal vector for each of the triangles, which make up the shaded picture tessellation for the component.
	IGetTessTriangleCount	Gets the number of triangles that make up the shaded picture tessellation for this component.
	IGetTessTriangles	Gets the triangles that make up the shaded picture tessellation for this component.
	IGetTessTriStripEdges	Gets the edge IDs for the triangle strips.
	IGetTessTriStripEdgeSize	Gets the number of tessellation triangle edges.
	IGetTessTriStripNorms	Gets the normal vector for each of the triangles, which make up the shaded picture tessellation for this component.
	IGetTessTriStrips	Gets the vertices that make up the shaded picture tessellation for this component.
	IGetTessTriStripSize	Gets the array size of floats required to contain the data returned when calling IComponent2::IGetTessTriStrips .
	IGetVisibility	Gets the visibility state for this component.
	IGetXform	Obsolete. Superseded by IComponent2::Transform2
	IListExternalFileReferences	Obsolete. Superseded by IComponent2::IListExternalFileReferences2
	IListExternalFileReferences2	Gets the names and statuses of the external references on the component.

		
	IRemoveMaterialProperty2	Removes material property values from the component.
	IsDisplayDataOutOfDate	Gets the status of the display data for this component.
	IsEnvelope	Gets whether this component is an envelope.
	ISetMaterialPropertyValues2	Sets the material properties for this component.
	ISetVisibility	Sets the visibility state for this component.
	ISetXform	Obsolete. Superseded by IComponent2::Transform2 .
	ISetXformAndSolve	Obsolete. Superseded by IComponent2::SetTransformAndSolve2 .
	IsFixed	Gets whether the component is fixed or floating.
	IsHidden	Gets whether this component is hidden or suppressed.
	IsLoaded	Gets whether a component is loaded.
	IsMirrored	Gets whether this component is mirrored.
	IsNameOverridden	Gets whether this component's name is overridden.
	IsPatternInstance	Gets whether the component is a member of a pattern instance.
	IsRoot	Gets whether this component is the root component.
	IsSmartComponent	Gets whether this component is a Smart Component.
	IsSuppressed	Gets whether this component is suppressed.
	ListExternalFileReferences	Obsolete. Superseded by IComponent2::ListExternalFileReferences2 .
	ListExternalFileReferences2	Gets the names and statuses of the external file references on the component.
	ListExternalFileReferencesCount	Gets the number of external references on the component.
	MakeVirtual	Obsolete. by Superseded by IComponent2::MakeVirtual2 .
	MakeVirtual2	Makes this component and optionally its child components virtual by saving them in the current assembly.
	RemoveMaterialProperty	Obsolete. Superseded by IComponent2::RemoveMaterialProperty2 .
	RemoveMaterialProperty2	Removes the appearance from the component.
	RemovePresentationTransform	Removes the presentation transform from this component.
	RemoveTexture	Removes the texture from this component in the specified configuration.
	RemoveTextureByDisplayState	Removes the texture applied to this component in the specified display state.
	ReplaceReference	Obsolete. Superseded by AssemblyDoc::ReplaceComponents .
	ResetPropertyExtension	Clears all of the values stored in the property extension.
	SaveVirtualComponent	Saves a virtual component to an external file.
	Select	Obsolete. Superseded by IComponent2::Select3 .
	Select2	Obsolete. Superseded by IComponent2::Select3 .
	Select3	Obsolete. Superseded by IComponent2::Select4 .
	Select4	Selects this component.
	SelectByMark	Obsolete. Superseded by IComponent2::Select3 .
	SetCosmosWorksMaterial	Assigns the material to use during analysis to this component.
	SetExcludeFromBOM2	Sets whether to exclude this component from the bills of materials (BOMs) in the specified configurations.
	SetMaterialIdName	Sets the material name for this component.
	SetMaterialPropertyValues2	Sets the material properties for this component.
	SetMaterialUserName	Sets the material user name for this component.
	SetReferencedDisplayStates	Sets the specified display state of this component to be referenced by the specified assembly display state(s).
	SetSmartComponentData	Sets the features, components, and feature references of a Smart Component.
	SetSuppression	Obsolete. Superseded by IComponent2::SetSuppression2 .

		
	SetSuppression2	Sets the suppression state of this component.
	SetTexture	Applies texture to this component in the specified configuration.
	SetTextureByDisplayState	Sets the texture applied to this component in the specified display state.
	SetTransformAndSolve	Obsolete. Superseded by IComponent2::SetTransformAndSolve2 .
	SetTransformAndSolve2	Obsolete. Superseded by IComponent2::SetTransformAndSolve3 .
	SetTransformAndSolve3	Sets the transform and solves for the mates for this component.
	SetVisibility	Sets the visibility state for this component.
	SetVisibilityInAsmDisplayStates	Sets the visibility of this component in the specified assembly display state(s).
	SetXform	Obsolete. Superseded by IComponent2::Transform2 .
	SetXformAndSolve	Obsolete. Superseded by IComponent2::SetTransformAndSolve2 .
	UpdateExternalFileReferences	Updates the external file references of this model.

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See Also

[IComponent2 Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IModelDoc2::GetPathName Method \(\)](#)

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IComponent2 Interface Methods

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : IComponent2 Interface

For a list of all members of this type, see [IComponent2 members](#).

Public Methods

Name	Description
 AddPropertyExtension	Adds a property extension to this component.
 DeSelect	Deselects this component.
 EnumBodies	Obsolete. Superseded by IComponent2::EnumBodies2 .
 EnumBodies2	Obsolete. Superseded by IComponent2::EnumBodies3 .
 EnumBodies3	Gets the bodies in the component in a multibody part.
 EnumRelatedBodies	Creates an enumerated list of bodies.
 EnumSectionedBodies	Gets the sectioned bodies seen in the specified view and returns them in an enumerated list.
 FeatureByName	Gets the specified feature for this component.
 FindAttribute	Finds an attribute on a component.
 FirstFeature	Gets the first feature in this component.
 GetBodies	Obsolete. Superseded by IComponent2::GetBodies2 .
 GetBodies2	Obsolete. Superseded by IComponent2::GetBodies3 .
 GetBodies3	Gets the bodies in this component.
 GetBody	Gets the body that belongs to this instance of this component.
 GetBox	Gets the bounding box for component.
 GetChildren	Gets all of the children components of this component.
 GetConfigurationCount	Gets the number of configurations for this lightweight assembly component.
 GetConfigurationNames	Gets the names of configurations for this lightweight assembly component.
 GetConstrainedStatus	Gets the constrained status of this component.

	GetCorresponding	Gets the corresponding object in the context of the assembly for a specific instance of the component.
	GetCorrespondingEntity	Gets the entity that corresponds with the Dispatch pointer in the context of the component.
	GetDecals	Gets the decals applied to this component.
	GetDecalsCount	Gets the number of decals applied to this component.
	GetDrawingComponent	Gets the drawing component for this component.
	GetExcludeFromBOM2	Gets whether this component is excluded from the bills of materials (BOMs) in the specified configurations.
	GetHiddenUnloadedChildrenCount	Gets the number of hidden children components of this component that were not loaded when an assembly was opened selectively .
	GetID	Gets the component ID for this component.
	GetImportedPath	Gets the full path name of the model imported for this component.
	GetMaterialIdName	Gets the material name for this component.
	GetMaterialPropertyValues2	Gets the material properties for this component.
	GetMaterialUserName	Gets the user-visible name of the material for this component.
	GetMates	Gets the mates for this component.
	GetModelDoc	Obsolete. Superseded by IComponent2::GetModelDoc2.
	GetModelDoc2	Gets the model document for this component.
	GetModelMaterialPropertyValues	Gets the material properties of this lightweight component in the specified configuration.
	GetModelTexture	Gets the texture applied to this lightweight component in the specified configuration.
	GetParent	Gets the parent component.
	GetPathName	Gets the full path name for this component.
	GetPLMID	Gets the ID of this SOLIDWORKS Design connected to the 3DEXPERIENCE platform document.
	GetPropertyExtension	Gets the property extension on this component.
	GetReferencedDisplayStates	Gets the display states of this component that are referenced by the specified assembly display state(s).
	GetRenderMaterials	Obsolete. Superseded by IComponent2::GetRenderMaterials2.
	GetRenderMaterials2	Gets the appearances applied to this component in the specified display states.
	GetRenderMaterialsCount	Obsolete. Superseded by IComponent2::GetRenderMaterialsCount2.
	GetRenderMaterialsCount2	Gets the number of appearances applied to this component in the specified display states.
	GetRepresentationParent	Gets the representation parent of this SOLIDWORKS Design connected to the 3DEXPERIENCE platform document.
	GetSectionedBodies	Gets the sectioned bodies in the specified section view.
	GetSelectByIdString	Gets the name of the component for possible use with IModelDocExtension::SelectById2 , for selectively opening a document using ISldWorks::OpenDoc7 and IDocumentSpecification , etc.
	GetSmartComponentData	Gets the features, components, and feature references of a Smart Component.
	GetSpecificTransform	Get the collapsed or exploded transform of a component when the assembly is exploded.
	GetSuppression	Obsolete. Superseded by IComponent2::GetSuppression2.
	GetSuppression2	Gets the suppression state of this component.
	GetTessNorms	Gets the normal vector for each of the triangles, which make up the shaded picture tessellation for the component.
	GetTessTriangles	Gets the triangles that make up the shaded picture tessellation for this component.
	GetTessTriStripEdges	Gets the edge IDs for the triangle strips.
	GetTessTriStripNorms	Gets the normal vector for each of the triangles, which make up the shaded picture tessellation for this component.
	GetTessTriStrips	Gets the vertices that make up the shaded picture tessellation for this component.
	GetTexture	Gets the texture applied to this component in the specified configuration.
	GetTotalTransform	Combines the original transform of this component with the presentation transform of this component.
	GetType	Gets this lightweight assembly component's document type.
	GetUnloadedComponentNames	Gets the component's unloaded children components' path names, referenced configuration names, reasons why they are unloaded, document types, and names.
	GetVisibility	Gets the visibility state for this component.

		
	GetVisibilityInAsmDisplayStates	Gets the visibilities of this component in the specified assembly display state(s).
	GetXform	Obsolete. Superseded by IComponent2::Transform2 .
	HasAssemblyFeatures	Gets whether this component has assembly features in the specified configuration.
	HasMaterialPropertyValues	Gets whether this component has an appearance.
	HasUnloadedComponents	Gets whether this component has hidden or suppressed unloaded children components.
	IFindAttribute	Finds an attribute on a component.
	IGetBody	Gets the body that belongs to this instance of this component.
	IGetBox	Gets the bounding box for component.
	IGetChildren	Gets all of the children components of this component.
	IGetChildrenCount	Gets the number of children components for this component.
	IGetCorrespondingEntity	Gets the entity that corresponds with the Dispatch pointer in the context of the component.
	IGetDecals	Gets the decals applied to this component.
	IGetMaterialPropertyValues2	Gets the material properties for this component.
	IGetMaterialPropertyValuesForFace	Gets the color of the specified face.
	IGetModelDoc	Obsolete. Superseded by IComponent2::GetModelDoc2 .
	IGetModelMaterialPropertyValues	Gets the material properties of this lightweight component in the specified configuration.
	IGetRenderMaterials	Obsolete. Superseded by IComponent2::GetRenderMaterials2 .
	IGetTemporaryBodyID	Gets the current body tag ID, which is not a permanent ID.
	IGetTessNorms	Gets the normal vector for each of the triangles, which make up the shaded picture tessellation for the component.
	IGetTessTriangleCount	Gets the number of triangles that make up the shaded picture tessellation for this component.
	IGetTessTriangles	Gets the triangles that make up the shaded picture tessellation for this component.
	IGetTessTriStripEdges	Gets the edge IDs for the triangle strips.
	IGetTessTriStripEdgeSize	Gets the number of tessellation triangle edges.
	IGetTessTriStripNorms	Gets the normal vector for each of the triangles, which make up the shaded picture tessellation for this component.
	IGetTessTriStrips	Gets the vertices that make up the shaded picture tessellation for this component.
	IGetTessTriStripSize	Gets the array size of floats required to contain the data returned when calling IComponent2::IGetTessTriStrips .
	IGetVisibility	Gets the visibility state for this component.
	IGetXform	Obsolete. Superseded by IComponent2::Transform2 .
	IListExternalFileReferences	Obsolete. Superseded by IComponent2::IListExternalFileReferences2 .
	IListExternalFileReferences2	Gets the names and statuses of the external references on the component.
	IRemoveMaterialProperty2	Removes material property values from the component.
	IsDisplayDataOutOfDate	Gets the status of the display data for this component.
	IsEnvelope	Gets whether this component is an envelope.
	ISetMaterialPropertyValues2	Sets the material properties for this component.
	ISetVisibility	Sets the visibility state for this component.
	ISetXform	Obsolete. Superseded by IComponent2::Transform2 .
	ISetXformAndSolve	Obsolete. Superseded by IComponent2::SetTransformAndSolve2 .
	IsFixed	Gets whether the component is fixed or floating.
	IsHidden	Gets whether this component is hidden or suppressed.
	IsLoaded	Gets whether a component is loaded.
	IsMirrored	Gets whether this component is mirrored.
	IsNameOverridden	Gets whether this component's name is overridden.

		
	IsPatternInstance	Gets whether the component is a member of a pattern instance.
	IsRoot	Gets whether this component is the root component.
	IsSmartComponent	Gets whether this component is a Smart Component.
	IsSuppressed	Gets whether this component is suppressed.
	ListExternalFileReferences	Obsolete. Superseded by IComponent2::ListExternalFileReferences2 .
	ListExternalFileReferences2	Gets the names and statuses of the external file references on the component.
	ListExternalFileReferencesCount	Gets the number of external references on the component.
	MakeVirtual	Obsolete. by Superseded by IComponent2::MakeVirtual2 .
	MakeVirtual2	Makes this component and optionally its child components virtual by saving them in the current assembly.
	RemoveMaterialProperty	Obsolete. Superseded by IComponent2::RemoveMaterialProperty2 .
	RemoveMaterialProperty2	Removes the appearance from the component.
	RemovePresentationTransform	Removes the presentation transform from this component.
	RemoveTexture	Removes the texture from this component in the specified configuration.
	RemoveTextureByDisplayState	Removes the texture applied to this component in the specified display state.
	ReplaceReference	Obsolete. Superseded by AssemblyDoc::ReplaceComponents .
	ResetPropertyExtension	Clears all of the values stored in the property extension.
	SaveVirtualComponent	Saves a virtual component to an external file.
	Select	Obsolete. Superseded by IComponent2::Select3 .
	Select2	Obsolete. Superseded by IComponent2::Select3 .
	Select3	Obsolete. Superseded by IComponent2::Select4 .
	Select4	Selects this component.
	SelectByMark	Obsolete. Superseded by IComponent2::Select3 .
	SetCosmosWorksMaterial	Assigns the material to use during analysis to this component.
	SetExcludeFromBOM2	Sets whether to exclude this component from the bills of materials (BOMs) in the specified configurations.
	SetMaterialIdName	Sets the material name for this component.
	SetMaterialPropertyValues2	Sets the material properties for this component.
	SetMaterialUserName	Sets the material user name for this component.
	SetReferencedDisplayStates	Sets the specified display state of this component to be referenced by the specified assembly display state(s).
	SetSmartComponentData	Sets the features, components, and feature references of a Smart Component.
	SetSuppression	Obsolete. Superseded by IComponent2::SetSuppression2 .
	SetSuppression2	Sets the suppression state of this component.
	SetTexture	Applies texture to this component in the specified configuration.
	SetTextureByDisplayState	Sets the texture applied to this component in the specified display state.
	SetTransformAndSolve	Obsolete. Superseded by IComponent2::SetTransformAndSolve2 .
	SetTransformAndSolve2	Obsolete. Superseded by IComponent2::SetTransformAndSolve3 .
	SetTransformAndSolve3	Sets the transform and solves for the mates for this component.
	SetVisibility	Sets the visibility state for this component.
	SetVisibilityInAsmDisplayStates	Sets the visibility of this component in the specified assembly display state(s).
	SetXform	Obsolete. Superseded by IComponent2::Transform2 .
	SetXformAndSolve	Obsolete. Superseded by IComponent2::SetTransformAndSolve2 .
	UpdateExternalFileReferences	Updates the external file references of this model.

See Also

[IComponent2 Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IModelDoc2::GetPathName Method \(\)](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IComponent2 Interface Properties

[See Also](#)

[SolidWorks.Interop.sldworks Namespace](#) : IComponent2 Interface

For a list of all members of this type, see [IComponent2 members](#).

Public Properties

Name	Description
 ComponentReference	Gets or sets a component reference for this component.
 CustomPropertyManager	Gets the configuration-specific custom property manager for this assembly component.
 DisplayTitle	Gets this component's title as displayed in the FeatureManager design tree.
 ExcludeFromBOM	Obsolete. Superseded by IComponent2::GetExcludeFromBOM2 and IComponent2::SetExcludeFromBOM2 .
 IMaterialPropertyValues	Gets or sets the material properties for the selected component in the active configuration.
 IsGraphicsOnly	Gets whether this component is graphics only.
 IsSpeedPak	Gets whether the active configuration for this component is SpeedPak.
 IsVirtual	Gets whether this component is a virtual component. NOTE: This property is a get-only property. Set is not implemented.
 MaterialPropertyValues	Gets or sets the material properties for the selected component in the active configuration.
 Name	Obsolete. Superseded by IComponent2::Name2 .
 Name2	Gets or sets the name of the selected component.
 PresentationTransform	Gets or sets the component transform.
 ReferencedConfiguration	Gets or sets the active configuration used by this component.
 ReferencedDisplayState	Obsolete. Superseded by IComponent2::ReferencedDisplayState2 .
 ReferencedDisplayState2	Gets or sets the active display state of this component.
 Solving	Gets the Solve as option (rigid or flexible) of this component.
 Transform	Obsolete. Superseded by IComponent2::Transform2 .
 Transform2	Gets or sets the component transform.
 UseNamedConfiguration	Gets whether a specified configuration or the in-use/last active configuration is used.
 Visible	Gets or sets the visibility state of this component.

[Top](#)

See Also

[IComponent2 Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IModelDoc2::GetPathName Method \(\)](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

AddPropertyExtension Method (IComponent2)

[See Also](#)

[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : AddPropertyExtension Method (IComponent2)

Adds a property extension to this component.

.NET Syntax

Visual Basic (Declaration)

```
Function AddPropertyExtension( _  
    ByVal PropertyExtension As System.Object _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim PropertyExtension As System.Object  
Dim value As System.Integer  
  
value = instance.AddPropertyExtension(PropertyExtension)
```

C#

```
System.int AddPropertyExtension(  
    System.object PropertyExtension  
)
```

C++/CLI

```
System.int AddPropertyExtension(  
& System.Object^ PropertyExtension  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PropertyExtension
Value of the property extension to add to the component

Return Value

1 if the property extension is added to the component, -1 if not

Visual Basic for Applications (VBA) Syntax

See [Component2::AddPropertyExtension](#).

Remarks

This method does not support adding multiple property extensions to the same component.

To use this method:

1. Declare the variable.
2. Assign the variable a value: float, integer, or string.
3. Call this method to add the value to the component.

NOTE: SOLIDWORKS recommends that you use the [IAttribute](#), [IAttributeDef](#), and [IParameter](#) interfaces instead of this method. These interface provide more flexibility.

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IComponent2::GetPropertyExtension Method](#)
[IComponent2::ResetPropertyExtension Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ComponentReference Property (IComponent2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : ComponentReference Property (IComponent2)

Gets or sets a component reference for this component.

.NET Syntax

```
Property ComponentReference As System.String
```



```
Dim instance As IComponent2
Dim value As System.String

instance.ComponentReference = value
value = instance.ComponentReference
```

```
System.string ComponentReference {get; set;}
```

```
property System.String^ ComponentReference {
    System.String^ get();
    void set ( & System.String^ value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Component reference string

Visual Basic for Applications (VBA) Syntax

See [Component2::ComponentReference](#).

Example

[Get Component Name from Selected Entity \(VB.NET\)](#)
[Get Component Name from Selected Entity \(VBA\)](#)
[Get Component Name from Selected Entity \(C#\)](#)

Remarks

Call [IModelDoc2::ForceRebuild3](#) after setting a component reference.

To remove a component reference, set the component reference string to "".

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

CustomPropertyManager Property (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : CustomPropertyManager Property (IComponent2)

Gets the configuration-specific custom property manager for this assembly component.

.NET Syntax

```
ReadOnly Property CustomPropertyManager( _
    ByVal ConfigurationName As System.String _
) As System.Object
```

```
Dim instance As IComponent2
Dim ConfigurationName As System.String
Dim value As System.Object

value = instance.CustomPropertyManager(ConfigurationName)
```

C#

```
System.Object CustomPropertyManager(
    System.String ConfigurationName
) {get;}
```

C++/CLI

```
property System.Object^ CustomPropertyManager {
    System.Object^ get(System.String^ ConfigurationName);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ConfigurationName

Name of the configuration for which to get custom properties; empty string to obtain document-specific custom properties

Property Value

[ICustomPropertyManager](#)

Visual Basic for Applications (VBA) Syntax

See [Component2::CustomPropertyManager](#).

Example

'VBA

'=====

'Preconditions:

- ' 1. Open *public_documents\SOLIDWORKS\SOLIDWORKS 2024\samples\tutorial\mbd\mbdadvanced\beater_assembly.sldasm*.
- ' 2. Create a configuration-specific custom property for **pan cross head<3>**.
 - ' a. Open **pan cross head<3>**.
 - ' b. Select the ConfigurationManager tab.
 - ' c. Right-click on CR-PHMS 0.073-72x0.125x0.125-N and select **Properties**.
 - ' d. In Configuration Properties, click **Custom Properties...**
 - ' e. On the Configuration Specific tab, create a custom property of type Text and provide a value.
- ' 2. Open the Immediate window.

'Postconditions: Displays the configuration-specific custom properties of all of the components in the assembly.

'=====

Option Explicit

```
Dim swApp As SldWorks.SldWorks
Dim swModelDoc As SldWorks.ModelDoc2
Dim swCompCur As SldWorks.Component2
Dim sCompName As String
Dim sRefConfigName As String
Dim swCompModelDoc As SldWorks.ModelDoc2
Dim swCompModelDocExt As SldWorks.ModelDocExtension
Dim swCompModelDocConf As SldWorks.Configuration
Dim swCompModelDocCustPropMgr As SldWorks.CustomPropertyManager
Dim swAssyDoc As SldWorks.AssemblyDoc
Dim vComps As Variant
Dim vComfigNames As Variant
Dim iCompIndex As Integer
Dim swCustPropMgrOfCurComp As SldWorks.CustomPropertyManager
Dim sConfigName As String
Dim iConfigIndex As Integer
```

```

Dim boolstatus As Boolean
Dim longstatus As Long, longwarnings As Long

Sub main()

    Set swApp = Application.SldWorks
    Set swModelDoc = swApp.ActiveDoc

    If swModelDoc.GetType = swDocASSEMBLY Then

        Set swAssyDoc = swModelDoc

        vComps = swAssyDoc.GetComponents(True)
        If Not IsEmpty(vComps) Then

            For iCompIndex = 0 To UBound(vComps)
                Set swCompCur = vComps(iCompIndex)
                sCompName = swCompCur.Name2
                Debug.Print ""
                Debug.Print "*****"
                Debug.Print "Component: " & sCompName

                sRefConfigName = swCompCur.ReferencedConfiguration
                Debug.Print "Referenced configuration: " & sRefConfigName

                'Get the component's document-level CustomPropertyManager
                Set swCustPropMgrOfCurComp = swCompCur.CustomPropertyManager("")
                DisplayCustomPropInfo "", swCustPropMgrOfCurComp

                Set swCompModelDoc = swCompCur.GetModelDoc2
                vComfigNames = swCompModelDoc.GetConfigurationNames

                'Get the configuration-specific CustomPropertyManager for each of the current component's configurations
                If Not IsEmpty(vComfigNames) Then
                    For iConfigIndex = 0 To UBound(vComfigNames)
                        sConfigName = vComfigNames(iConfigIndex)
                        Set swCustPropMgrOfCurComp = swCompCur.CustomPropertyManager(sConfigName)
                        DisplayCustomPropInfo sConfigName, swCustPropMgrOfCurComp
                    Next iConfigIndex
                End If
            Next iCompIndex

            Debug.Print ""
        End If
    End If
End Sub

Sub DisplayCustomPropInfo(sConfigName As String, swCustPropMgr As CustomPropertyManager)
    Dim CustPropCount As Long
    Dim sValOut As String
    Dim sResValOut As String
    Dim bWasResolved As Boolean
    Dim bLinkToProperty As Boolean
    Dim IResult As Long
    Dim vNames As Variant
    Dim i As Long
    Dim IValue As Long
    Dim bValue As Boolean
    Dim ILinkPropValue As Long
    Dim ISetValue As Long
    Dim PropNames As Variant
    Dim PropTypes As Variant
    Dim PropValues As Variant
    Dim Resolved As Variant
    Dim PropLink As Variant
    Dim IGetAllValue As Long
    Dim IPropIndex As Long

    Debug.Print ""
    If sConfigName = "" Then
        Debug.Print "Custom properties: Document level"
    Else
        Debug.Print "Custom properties: Configuration = " & sConfigName
    End If

    If swCustPropMgr Is Nothing Then
        Debug.Print " ***** CustomPropertyManager() Failed! *****"
    Else

        CustPropCount = swCustPropMgr.Count
        Debug.Print " Count: " & CustPropCount
    End If
End Sub

```

```

If CustPropCount > 0 Then

    vNames = swCustPropMgr.GetNames

    For i = 0 To UBound(vNames)
        Debug.Print " " & vNames(i)

        IResult = swCustPropMgr.Get6(vNames(i), False, sValOut, sResValOut, bWasResolved, bLinkToProperty)

        Debug.Print "    Value: " & sResValOut & "    Evaluated value: " & sValOut

        IValue = swCustPropMgr.GetType2(vNames(i))
        Select Case IValue
            Case swCustomInfoUnknown
                Debug.Print "    Type: swCustomInfoUnknown"
            Case swCustomInfoNumber
                Debug.Print "    Type: swCustomInfoNumber"
            Case swCustomInfoDouble
                Debug.Print "    Type: swCustomInfoDouble"
            Case swCustomInfoYesOrNo
                Debug.Print "    Type: swCustomInfoYesOrNo"
            Case swCustomInfoText
                Debug.Print "    Type: swCustomInfoText"
            Case swCustomInfoDate
                Debug.Print "    Type: swCustomInfoDate"
            Case Else
                Debug.Print "    Type: Value not in enum!"
        End Select

        bValue = swCustPropMgr.IsCustomPropertyEditable(vNames(i), sConfigName)
        Debug.Print "    Editable: " & bValue

        ILinkPropValue = swCustPropMgr.LinkProperty(vNames(i), True)
        Select Case ILinkPropValue
            Case swCustomLinkSetResult_OK
                Debug.Print "    LinkProperty(True): swCustomLinkSetResult_OK"
            Case swCustomLinkSetResult_NotPresent
                Debug.Print "    LinkProperty(True): swCustomLinkSetResult_NotPresent"
            Case swCustomLinkSetResult_Legacy
                Debug.Print "    LinkProperty(True): swCustomLinkSetResult_Legacy"
            Case swCustomLinkSetResult_UserProp
                Debug.Print "    LinkProperty(True): swCustomLinkSetResult_UserProp"
            Case Else
                Debug.Print "    Type: Value not in enum!"
        End Select

        ILinkPropValue = swCustPropMgr.LinkProperty(vNames(i), False)
        Select Case ILinkPropValue
            Case swCustomLinkSetResult_OK
                Debug.Print "    LinkProperty(False): swCustomLinkSetResult_OK"
            Case swCustomLinkSetResult_NotPresent
                Debug.Print "    LinkProperty(False): swCustomLinkSetResult_NotPresent"
            Case swCustomLinkSetResult_Legacy
                Debug.Print "    LinkProperty(False): swCustomLinkSetResult_Legacy"
            Case swCustomLinkSetResult_UserProp
                Debug.Print "    LinkProperty(False): swCustomLinkSetResult_UserProp"
            Case Else
                Debug.Print "    Type: Value not in enum!"
        End Select

    Next i

    IGetAllValue = swCustPropMgr.GetAll3(PropNames, PropTypes, PropValues, Resolved, PropLink)
    For IPropIndex = 0 To UBound(PropNames)
        Debug.Print "    " & PropNames(IPropIndex) & ", " & PropTypes(IPropIndex) & ", " & PropValues(IPropIndex) & ", " &
        Resolved(IPropIndex) & ", " & PropLink(IPropIndex)
    Next IPropIndex

End If
End If
End Sub

```

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)

Availability

SOLIDWORKS 2024 FCS, Revision Number 32

SOLIDWORKS API Help

[Send comments on this topic.](#)

DeSelect Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : DeSelect Method (IComponent2)

Deselects this component.

.NET Syntax

Visual Basic (Declaration)

```
Function DeSelect() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim value As System.Boolean  
  
value = instance.DeSelect()
```

C#

```
System.bool DeSelect()
```

C++/CLI

```
System.bool DeSelect();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if the component was successfully deselected, false if not

Visual Basic for Applications (VBA) Syntax

See [Component2::DeSelect](#).

Remarks

Use [IModelDoc2::DeSelectByID](#) instead of this method. This method does not work well when a PropertyManager page is open or a command is running. IModelDoc2::DeSelectByID handles selection correctly whether or not a command is running.

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IComponent2::Select3 Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

DisplayTitle Property (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : DisplayTitle Property (IComponent2)

Gets this component's title as displayed in the FeatureManager design tree.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property DisplayTitle As System.String
```

Visual Basic (Usage)



```
Dim instance As IComponent2
Dim value As System.String

value = instance.DisplayTitle
```

C#

```
System.string DisplayTitle {get;}
```

C++/CLI

```
property System.String^ DisplayTitle {
    System.String^ get();
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Display title

Visual Basic for Applications (VBA) Syntax

See [Component2::DisplayTitle](#).

Remarks

To get or set the display title of an assembly, drawing or part, use [IModelDocExtension::DisplayTitle](#).

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)

Availability

SOLIDWORKS 2021 SP03, Revision Number 29.3

SOLIDWORKS API Help

[Send comments](#) on this topic.

EnumBodies Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : EnumBodies Method (IComponent2)

Obsolete. Superseded by [IComponent2::EnumBodies2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function EnumBodies( _
    ByVal BodyType As System.Integer _
) As EnumBodies2
```

Visual Basic (Usage)



```
Dim instance As IComponent2
Dim BodyType As System.Integer
Dim value As EnumBodies2

value = instance.EnumBodies(BodyType)
```

C#

```
EnumBodies2 EnumBodies(  
    System.int BodyType  
)
```

C++/CLI

```
EnumBodies2^ EnumBodies(  
&    System.int BodyType  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

BodyType

Visual Basic for Applications (VBA) Syntax

See [Component2::EnumBodies](#).

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

EnumBodies2 Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : EnumBodies2 Method (IComponent2)

Obsolete. Superseded by [IComponent2::EnumBodies3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function EnumBodies2( _  
    ByVal BodyType As System.Integer _  
) As EnumBodies2
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim BodyType As System.Integer  
Dim value As EnumBodies2  
  
value = instance.EnumBodies2(BodyType)
```

C#

```
EnumBodies2 EnumBodies2(  
    System.int BodyType  
)
```

C++/CLI

```
EnumBodies2^ EnumBodies2(  
&    System.int BodyType  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

BodyType

Type of body as defined by [swBodyType_e](#)

Return Value

[Enumerated list of bodies](#)

Visual Basic for Applications (VBA) Syntax

See [Component2::EnumBodies2](#).

Remarks

This method only supports solid and sheet (surface) body types.

The difference between this method and the now obsolete `IComponent2::EnumBodies` is that the new method supports lightweight components. The now obsolete `IComponent2::EnumBodies` does not.

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IComponent2::GetBodies3 Method](#)
[IComponent2::GetBody Method](#)
[IComponent2::IGetBody Method](#)

Availability

SOLIDWORKS 2006 SP2, Revision Number 14.2

SOLIDWORKS API Help

[Send comments on this topic.](#)

EnumBodies3 Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : EnumBodies3 Method (IComponent2)

Gets the bodies in the component in a multibody part.

.NET Syntax

Visual Basic (Declaration)

```
Function EnumBodies3( _  
    ByVal BodyType As System.Integer, _  
    ByVal VisibleOnly As System.Boolean _  
) As EnumBodies2
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim BodyType As System.Integer  
Dim VisibleOnly As System.Boolean  
Dim value As EnumBodies2  
  
value = instance.EnumBodies3(BodyType, VisibleOnly)
```

C#

```
EnumBodies2 EnumBodies3(  
    System.int BodyType,  
    System.bool VisibleOnly  
)
```

C++/CLI

```
EnumBodies2^ EnumBodies3(  
&    System.int BodyType,  
&    System.bool VisibleOnly  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

BodyType

Type of body as defined by [swBodyType_e](#)

VisibleOnly

True to get visible bodies only, false to get all bodies

Return Value

[Enumerated list of bodies](#)

Visual Basic for Applications (VBA) Syntax

See [Component2::EnumBodies3](#).

Remarks

This method only supports solid and sheet (surface) types.

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IComponent2::EnumRelatedBodies Method \(\)](#)
[IComponent2::EnumSectionedBodies Method \(\)](#)
[IComponent2::GetBodies3 Method \(\)](#)
[IComponent2::GetBody Method \(\)](#)
[IComponent2::IGetBody Method \(\)](#)
[IPartDoc::EnumBodies3 Method \(\)](#)

Availability

SOLIDWORKS 2016 FCS, Revision Number 24.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

EnumRelatedBodies Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : EnumRelatedBodies Method (IComponent2)

Creates an enumerated list of bodies.

.NET Syntax

Visual Basic (Declaration)

```
Function EnumRelatedBodies() As EnumBodies2
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim value As EnumBodies2
```

```
value = instance.EnumRelatedBodies()
```

C#

```
EnumBodies2 EnumRelatedBodies()
```

C++/CLI

```
EnumBodies2^ EnumRelatedBodies();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Pointer to the [enumerated list of bodies](#)

Visual Basic for Applications (VBA) Syntax

See [Component2::EnumRelatedBodies](#).

Remarks

The list contains only those bodies associated with reference surfaces. You get a list of bodies that are related to the model, but the list does not include the part body itself.

A reference surface feature might consist of one or more surfaces sewn together. SOLIDWORKS represents each reference surface feature with two bodies: one to represent the front faces and one to represent the back faces. Each IBody2 object has one or more faces depending on whether the reference surface feature is a set of sewn surfaces or a single underlying surface. The corresponding faces for each body pair should have normal vectors that are opposite.

To use the enumerated list, see [IEnumBodies2::Next](#), [IEnumBodies2::Skip](#), [IEnumBodies2::Reset](#), and [IEnumBodies2::Clone](#).

If a component is suppressed or lightweight, this method might return NULL because the component has not been loaded into memory by SOLIDWORKS. For more information on lightweight components, see [Work With Lightweight Components](#).

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

EnumSectionedBodies Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : EnumSectionedBodies Method (IComponent2)

Gets the sectioned bodies seen in the specified view and returns them in an enumerated list.

.NET Syntax

Visual Basic (Declaration)

```
Function EnumSectionedBodies( _  
    ByVal ViewIn As ModelView _  
) As EnumBodies2
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim ViewIn As ModelView  
Dim value As EnumBodies2  
  
value = instance.EnumSectionedBodies(ViewIn)
```

C#

```
EnumBodies2 EnumSectionedBodies(  
    ModelView ViewIn  
)
```

C++/CLI

```
EnumBodies2^ EnumSectionedBodies(  
& ModelView^ ViewIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ViewIn

Pointer to the [model view](#) displaying the sectioned view

Return Value

Pointer to the [enumerated list of bodies](#)

Visual Basic for Applications (VBA) Syntax

See [Component2::EnumSectionedBodies](#).

Remarks

To determine if the model view you want is currently displaying a sectioned view, use [IModelView::GetDisplayState](#).

If you need the full body representation, use [IComponent2::GetBody](#), [IPartDoc::GetBodies2](#), or [IPartDoc::EnumBodies3](#), which ignore the sectioning operation.

For Dispatch implementations, see [IComponent2::GetSectionedBodies](#).

If a component is suppressed or lightweight, this method returns NULL because the component is not loaded into memory by SOLIDWORKS. For more information on lightweight components, see [Work With Lightweight Components](#).

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ExcludeFromBOM Property (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : ExcludeFromBOM Property (IComponent2)

Obsolete. Superseded by [IComponent2::GetExcludeFromBOM2](#) and [IComponent2::SetExcludeFromBOM2](#).

.NET Syntax

Visual Basic (Declaration)

Property ExcludeFromBOM As System.Boolean

Visual Basic (Usage)



```
Dim instance As IComponent2
Dim value As System.Boolean

instance.ExcludeFromBOM = value

value = instance.ExcludeFromBOM
```

C#

```
System.bool ExcludeFromBOM {get; set;}
```

C++/CLI

```
property System.bool ExcludeFromBOM {
    System.bool get();
    void set ( & System.bool value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to exclude this component from the BOM, false to include it

Visual Basic for Applications (VBA) Syntax

See [Component2::ExcludeFromBOM](#).

Remarks

This property applies to [table-based bills of materials](#); it does not apply to Microsoft Excel-based bills of materials.

If you change this property to true, the name of the component in the FeatureManager design tree changes; **(Excluded from BOM)** is appended. To update the FeatureManager design tree, call [IFeatureManager::UpdateFeatureTree](#).

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IAssemblyDoc::CompConfigProperties5 Method \(\)](#)

Availability

SOLIDWORKS 2005 SP1, Revision Number 13.1

SOLIDWORKS API Help

[Send comments on this topic.](#)

FeatureByName Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : FeatureByName Method (IComponent2)

Gets the specified feature for this component.

.NET Syntax

Visual Basic (Declaration)

```
Function FeatureByName( _  
    ByVal Name As System.String _  
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim Name As System.String  
Dim value As Feature  
  
value = instance.FeatureByName(Name)
```

C#

```
Feature FeatureByName(  
    System.string Name  
)
```

C++/CLI

```
Feature^ FeatureByName(  
& System.String^ Name  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Name
Name of the feature to retrieve

Return Value

Pointer to the [IFeature](#) object

Visual Basic for Applications (VBA) Syntax

See [Component2::FeatureByName](#).

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)

Availability

FindAttribute Method (IComponent2)

[See Also](#)[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : FindAttribute Method (IComponent2)

Finds an attribute on a component.

.NET Syntax

Visual Basic (Declaration)

```
Function FindAttribute( _
    ByVal AttributeDef As System.Object, _
    ByVal WhichOne As System.Integer _
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IComponent2
Dim AttributeDef As System.Object
Dim WhichOne As System.Integer
Dim value As System.Object

value = instance.FindAttribute(AttributeDef, WhichOne)
```

C#

```
System.object FindAttribute(
    System.object AttributeDef,
    System.int WhichOne
)
```

C++/CLI

```
System.Object^ FindAttribute(
    & System.Object^ AttributeDef,
    & System.int WhichOne
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

AttributeDef

Attribute definition that you are looking for on this component

WhichOne

Index number of the attribute that you want to return; there might be several attributes on this component

Return Value

Pointer to the [attribute](#)

Visual Basic for Applications (VBA) Syntax

See [Component2::FindAttribute](#).

See Also

[IComponent2 Interface](#)[IComponent2 Members](#)[IComponent2::FindAttribute Method](#)

Availability

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : FirstFeature Method (IComponent2)

Gets the first feature in this component.

.NET Syntax

Visual Basic (Declaration)

```
Function FirstFeature() As Feature
```

Visual Basic (Usage)



```
Dim instance As IComponent2
Dim value As Feature

value = instance.FirstFeature()
```

C#

```
Feature FirstFeature()
```

C++/CLI

```
Feature^ FirstFeature();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Pointer to the first [feature](#) in this component

Visual Basic for Applications (VBA) Syntax

See [Component2::FirstFeature](#).

Example

[Select Origin of Assembly Component \(VBA\)](#)
[Traverse Assembly and Hide All Sketches \(VBA\)](#)
[Traverse Assembly at Component and Feature Levels \(VBA\)](#)
[Traverse Assembly at Component and Feature Levels Using Recursion \(VBA\)](#)
[Traverse Assembly at Component and Feature Levels Using Recursion \(VB.NET\)](#)
[Traverse Assembly at Component and Feature Levels Using Recursion \(C#\)](#)

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IFeature::GetNextFeature Method](#)
[IFeature::GetNextSubFeature Method](#)
[IFeature::IGetNextFeature Method](#)
[IFeature::IGetNextFeature Method](#)
[IModelDoc2::FirstFeature Method](#)
[IModelDoc2::IFirstFeature Method](#)
[IPartDoc::FirstFeature Method](#)
[IPartDoc::IFirstFeature Method](#)

Availability

SOLIDWORKS 2001Plus SP5, Revision Number 10.5

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetBodies Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : GetBodies Method (IComponent2)

Obsolete. Superseded by [IComponent2::GetBodies2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetBodies( _  
    ByVal BodyType As System.Integer _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim BodyType As System.Integer  
Dim value As System.Object  
  
value = instance.GetBodies(BodyType)
```

C#

```
System.object GetBodies(  
    System.int BodyType  
)
```

C++/CLI

```
System.Object^ GetBodies(  
& System.int BodyType  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

BodyType

Visual Basic for Applications (VBA) Syntax

See [Component2::GetBodies](#).

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)

Availability

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetBodies2 Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : GetBodies2 Method (IComponent2)

Obsolete. Superseded by [IComponent2::GetBodies3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetBodies2( _  
    ByVal BodyType As System.Integer _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim BodyType As System.Integer  
Dim value As System.Object  
  
value = instance.GetBodies2(BodyType)
```

C#

```
System.Object GetBodies2(  
    System.Int BodyType  
)
```

C++/CLI

```
System.Object^ GetBodies2(  
& System.Int BodyType  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

BodyType

Type of body as defined by [swBodyType_e](#)

Return Value

Array of [bodies](#) in the component

Visual Basic for Applications (VBA) Syntax

See [Component2::GetBodies2](#).

Remarks

This method only supports solid and sheet (surface) types and part components.

The difference between this method and the now obsolete `IComponent2::GetBodies` is that the new method supports lightweight components. The now obsolete `IComponent2::GetBodies` does not.

See Also

[IComponent2 Interface](#)

[IComponent2 Members](#)

Availability

SOLIDWORKS 2006 SP2, Revision Number 14.2

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetBodies3 Method (IComponent2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : GetBodies3 Method (IComponent2)

Gets the bodies in this component.

.NET Syntax

Visual Basic (Declaration)

```
Function GetBodies3( _  
    ByVal BodyType As System.Integer, _  
    ByRef BodiesInfo As System.Object _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim BodyType As System.Integer  
Dim BodiesInfo As System.Object  
Dim value As System.Object
```

```
value = instance.GetBodies3(BodyType, BodiesInfo)
```

C#

```
System.Object GetBodies3(  
    System.Int BodyType,  
    out System.Object BodiesInfo  
)
```

C++/CLI

```
System.Object^ GetBodies3(  
& System.Int BodyType,  
& [Out] System.Object^ BodiesInfo  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

BodyType

Type of body as defined by [swBodyType_e](#)

BodiesInfo

Array of information about the returned bodies as defined in [swBodyInfo_e](#)

Return Value

Array of [bodies](#) in the component

Visual Basic for Applications (VBA) Syntax

See [Component2::GetBodies3](#).

Example

Also see the [IView::GetVisibleDrawingComponents](#) examples.

Example

[Get Bodies in Components \(C#\)](#)
[Get Bodies in Components \(VB.NET\)](#)
[Get Bodies in Components \(VBA\)](#)
[Get Bodies in Components \(C++\)](#)

Remarks

This method:

- Only supports solid and surface bodies and part components,
- May vary the order in which bodies are returned,
- Supports lightweight components (the now obsolete IComponent2::GetBodies does not),
- and -
- Returns an array in BodiesInfo containing information about bodies that indicates whether they are normal or user bodies. User bodies are original component bodies that have been modified in an assembly (e.g., a normal component body is cut in the assembly, resulting in two user component bodies). User bodies are not created for surface bodies.

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IComponent2::EnumBodies2 Method](#)
[IComponent2::GetBody Method](#)
[IComponent2::IGetBody Method](#)
[IPartDoc::GetBodies2 Method](#)
[IBody2::IsSheetMetal Method \(\)](#)

Availability

SOLIDWORKS 2009 SP4, Revision Number 17.4

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetBody Method (IComponent2)

[See Also](#) [Example](#)



Gets the body that belongs to this instance of this component.

.NET Syntax

Visual Basic (Declaration)

```
Function GetBody() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IComponent2
Dim value As System.Object

value = instance.GetBody()
```

C#

```
System.object GetBody()
```

C++/CLI

```
System.Object^ GetBody();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Body

Visual Basic for Applications (VBA) Syntax

See [Component2::GetBody](#).

Example

[Combine Assembly Components into Part \(VBA\)](#)
[Get Mass Properties of Assembly Component \(VBA\)](#)
[Get Material Properties \(VBA\)](#)
[Select Component Face By Name \(VBA\)](#)
[Display Temporary Body \(C#\)](#)
[Display Temporary Body \(VB.NET\)](#)
[Display Temporary Body \(VBA\)](#)

Remarks

This method returns a valid IBody2 object only for fully resolved components that reference an [IPartDoc](#). For the root component, lightweight components or components that reference an AssemblyDoc, this method returns a NULL body object. However, if a component is in a multibody part, then this method returns the first body.

For COM applications, this method returns the status code ITF_E_COMPONENTNOTRESOLVED.

This method is different from the IPartDoc::Body method in that it recognizes assembly-level features and returns that information based on the component instance. IPartDoc::Body never recognizes assembly-level features because the feature information is kept with the assembly, not propagated down to the part file.

For example, if PartABC is added to an assembly twice, then changes to that IPartDoc object would affect both instances of the assembly IComponent2. Likewise, querying information from that IPartDoc object will not recognize changes in the assembly that may have altered only one of the components (for example, an assembly-level feature was added to one of the components). However, the component object recognizes the two instances of PartABC as two distinct IComponent2 objects and returns its information from the assembly level.

For example, one assembly configuration might have an assembly-level feature that cuts a hole through each of the components in the assembly. Using this method on each of the assembly components returns the body of each component with the hole feature that was applied in this particular configuration. If you switch to the configuration without the assembly-level hole and re-traverse the IComponent2 objects, then calling this method for each component returns the IBody2 object without the hole feature because it was applied in the other configuration.

For more information on lightweight components, see [IComponent2::GetSuppression](#), [IComponent2::SetSuppression2](#), and [IAssemblyDoc::ResolveAllLightWeightComponents](#).

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)

[IComponent2::IGetBody Method](#)
[IComponent2::EnumBodies2 Method](#)
[IComponent2::GetBodies3 Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetBox Method (IComponent2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : GetBox Method (IComponent2)

Gets the bounding box for component.

.NET Syntax

Visual Basic (Declaration)

```
Function GetBox( _  
    ByVal IncludeRefPlanes As System.Boolean, _  
    ByVal IncludeSketches As System.Boolean _  
    ) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim IncludeRefPlanes As System.Boolean  
Dim IncludeSketches As System.Boolean  
Dim value As System.Object  
  
value = instance.GetBox(IncludeRefPlanes, IncludeSketches)
```

C#

```
System.object GetBox(  
    System.bool IncludeRefPlanes,  
    System.bool IncludeSketches  
)
```

C++/CLI

```
System.Object^ GetBox(  
& System.bool IncludeRefPlanes,  
& System.bool IncludeSketches  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

IncludeRefPlanes

True includes reference planes with the returned bounding box, false does not

IncludeSketches

True includes sketches with the returned bounding box, false does not

Return Value

Object containing the two diagonal points that bound the component; the array contains 6 doubles

Visual Basic for Applications (VBA) Syntax

See [Component2::GetBox](#).

Example

[Get Assembly Bounding Box Using Components \(VBA\)](#)
[Get Assembly Bounding Box Using Components \(C#\)](#)
[Get Assembly Bounding Box Using Components \(VB.NET\)](#)

Remarks

The resulting box encloses the object, but it might not be the tightest box.

The X, Y, Z points returned by SOLIDWORKS are the lower- and upper-diagonal corners that bound the component with the box sides parallel to the X, Y and Z axes. SOLIDWORKS returns box dimensions that enclose the component and are typically close to the minimum possible size.

The return value is an array of doubles as follows:

[*XCorner1*, *YCorner1*, *ZCorner1*, *XCorner2*, *YCorner2*, *ZCorner2*]

It is possible for this method to return a NULL VARIANT for Dispatch. This occurs if your application calls IComponent2::GetBox with a component that represented a subassembly and that subassembly is not loaded.

The user interface behavior is the same. When the user selects a subassembly that is not loaded, there is no selection box around the subassembly. However, once the subassembly is loaded, there is a selection box.

IMPORTANT: The values returned are approximate and should not be used for comparison or calculation purposes. Furthermore, the bounding box may vary after rebuilding the model.

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IComponent2::IGetBox Method](#)
[IFeature::GetBox Method](#)
[IFeature::IGetBox Method](#)
[IFace2::GetBox Method](#)
[IFace2::IGetBox Method](#)
[IBody2::GetBodyBox Method](#)
[IBody2::IGetBodyBox Method](#)
[IAssemblyDoc::GetBox Method](#)
[IAssemblyDoc::IGetBox Method](#)
[IModelDocExtension::GetVisibleBox Method](#)
[IModelDocExtension::RemoveVisibleBox Method](#)
[IModelDocExtension::SetVisibleBox Method](#)
[IPartDoc::GetPartBox Method \(\)](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetChildren Method (IComponent2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : GetChildren Method (IComponent2)

Gets all of the children components of this component.

.NET Syntax

Visual Basic (Declaration)

```
Function GetChildren() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim value As System.Object  
  
value = instance.GetChildren()
```

C#

```
System.object GetChildren()
```

C++/CLI

```
System.Object^ GetChildren();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of [components](#)

Visual Basic for Applications (VBA) Syntax

See [Component2::GetChildren](#).

Example

[Get Display Dimensions, GTols, and Surface-finish Symbols \(VBA\)](#)
[Get Names of Bodies in Multibody Part \(VBA\)](#)
[Get Transforms of Assembly Components \(VBA\)](#)
[Make All Assembly Components Visible \(VBA\)](#)
[Traverse Assembly at Component and Feature Levels \(VBA\)](#)
[Traverse Assembly at Component and Feature Levels Using Recursion \(VBA\)](#)
[Traverse Assembly at Component and Feature Levels Using Recursion \(VB.NET\)](#)
[Traverse Assembly at Component and Feature Levels Using Recursion \(C#\)](#)

Remarks

If this assembly component is a part document, SOLIDWORKS returns NULL. If this assembly component is the root component or a subassembly, then this method returns the child components that belong to the assembly document.

The typical order of calls needed in assembly traversal is:

1. [IConfigurationManager.ActiveConfiguration](#) (called only once)
2. [IConfiguration::GetRootComponent](#) (called only once)
3. [IComponent2::GetChildren](#) (called recursively)

You must call the [IComponent2::GetChildren](#) method recursively because it returns only the immediate (one level) children. It does not get child components of the sub-assemblies. For example, if one of the child components of a component is a sub-assembly that has its own children, those children are not returned by this method. You need to call this method again from that sub-assembly component to get its children.

For a given component, this method returns all of the immediate child components. This includes suppressed, hidden, and lightweight components. Use [IComponent2::Visible](#) and [IComponent2::GetSuppression](#) to detect the component states.

Results of an assembly traversal vary based on the configuration currently displayed in your main assembly and based on the configuration referenced by the subassembly component. The list of child components that this method returns can be different depending on which configuration is referenced by the component (see [IConfigurationManager.ActiveConfiguration](#) and [IComponent2::ReferencedConfiguration](#)).

For example, if one configuration of your main assembly contains a suppressed subassembly, [IComponent2::GetChildren](#) returns an empty array when you call it from that suppressed subassembly component. As another example, a subassembly document (.sldasm file) can contain several configurations, each of which has varying states of suppression for its child components. When inserted into your main assembly, this subassembly document can reference any of these configurations. As a result, you might find that the child component suppression states vary based on which configuration is referenced by the subassembly component.

NOTE: Components might not be returned in the same order from call to call.

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[Component2::IGetChildren Method](#)
[IAssemblyDoc::GetComponentCount Method](#)
[IAssemblyDoc::GetComponents Method](#)
[IComponent2::GetParent Method](#)
[IConfiguration::GetRootComponent3 Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetConfigurationCount Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : [GetConfigurationCount Method \(IComponent2\)](#)

Gets the number of configurations for this lightweight assembly component.

.NET Syntax

Visual Basic (Declaration)

Function GetConfigurationCount() As System.Integer

Visual Basic (Usage)



```
Dim instance As IComponent2
Dim value As System.Integer

value = instance.GetConfigurationCount()
```

C#

```
System.int GetConfigurationCount()
```

C++/CLI

```
System.int GetConfigurationCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of configurations for this lightweight assembly component

Visual Basic for Applications (VBA) Syntax

See [Component2::GetConfigurationCount](#).

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)

Availability

SOLIDWORKS 2024 SP01, Revision Number 32.1

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetConfigurationNames Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : GetConfigurationNames Method (IComponent2)

Gets the names of configurations for this lightweight assembly component.

.NET Syntax

Visual Basic (Declaration)

```
Function GetConfigurationNames() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IComponent2
Dim value As System.Object

value = instance.GetConfigurationNames()
```

C#

```
System.object GetConfigurationNames()
```

C++/CLI

```
System.Object^ GetConfigurationNames();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of configuration names

Visual Basic for Applications (VBA) Syntax

See [Component2::GetConfigurationNames](#).

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)

Availability

SOLIDWORKS 2024 SP01, Revision Number 32.1

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetConstrainedStatus Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : GetConstrainedStatus Method (IComponent2)

Gets the constrained status of this component.

.NET Syntax

Visual Basic (Declaration)

```
Function GetConstrainedStatus() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim value As System.Integer  
  
value = instance.GetConstrainedStatus()
```

C#

```
System.int GetConstrainedStatus()
```

C++/CLI

```
System.int GetConstrainedStatus();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Constrained status as defined in [swConstrainedStatus_e](#)

Visual Basic for Applications (VBA) Syntax

See [Component2::GetConstrainedStatus](#).

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetCorresponding Method (IComponent2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : GetCorresponding Method (IComponent2)

Gets the corresponding object in the context of the assembly for a specific instance of the component.

.NET Syntax

Visual Basic (Declaration)

```
Function GetCorresponding( _  
    ByVal InputObject As System.Object _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim InputObject As System.Object  
Dim value As System.Object  
  
value = instance.GetCorresponding(InputObject)
```

C#

```
System.object GetCorresponding(  
    System.object InputObject  
)
```

C++/CLI

```
System.Object^ GetCorresponding(  
& System.Object^ InputObject  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

InputObject
Pointer to the Dispatch object

Return Value

Pointer to the corresponding object in the context of the assembly for an instance of the component

Visual Basic for Applications (VBA) Syntax

See [Component2::GetCorresponding](#).

Example

[Get Corresponding Objects in Assembly Component \(C#\)](#)
[Get Corresponding Objects in Assembly Component \(VB.NET\)](#)
[Get Corresponding Objects in Assembly Component \(VBA\)](#)

Remarks

Given an object in an underlying component document, this method gets the corresponding object in the context of the assembly for a specific instance of that component document (i.e., there can be more than one instance).

You can use this method with any object assigned a persistent reference or object ID; for example, [IAnnotation](#), [IFeature](#), [ISketch](#), [ISketchSegment](#), etc.

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IComponent2::GetCorrespondingEntity Method](#)
[IComponent2::IGetCorrespondingEntity Method](#)
[IModelDocExtension::GetCorresponding Method](#)
[IModelDocExtension::GetCorrespondingEntity Method](#)

Availability

SOLIDWORKS 2006 FCS, Revision Number 14.0



Gets the entity that corresponds with the Dispatch pointer in the context of the component.

.NET Syntax

Visual Basic (Declaration)

```
Function GetCorrespondingEntity( _  
    ByVal Entity As System.Object _  
) As System.Object
```

Visual Basic (Usage)

```
Dim instance As IComponent2  
Dim Entity As System.Object  
Dim value As System.Object  
  
value = instance.GetCorrespondingEntity(Entity)
```

C#

```
System.object GetCorrespondingEntity(  
    System.object Entity  
)
```

C++/CLI

```
System.Object^ GetCorrespondingEntity(  
& System.Object^ Entity  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Entity

Pointer to the Dispatch object of an entity (vertex, face, or edge)

Return Value

Pointer to the corresponding object in the context of the component

Visual Basic for Applications (VBA) Syntax

See [Component2::GetCorrespondingEntity](#).

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IComponent2::GetCorresponding_Method](#)
[IComponent2::IGetCorrespondingEntity_Method](#)
[IModelDocExtension::GetCorresponding_Method](#)
[IModelDocExtension::GetCorrespondingEntity_Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetDecals Method (IComponent2)

[See Also](#)



SolidWorks.Interop.sldworks Namespace > IComponent2 Interface : GetDecals Method (IComponent2)

Gets the decals applied to this component.

.NET Syntax

Visual Basic (Declaration)

```
Function GetDecals() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IComponent2
Dim value As System.Object

value = instance.GetDecals()
```

C#

```
System.Object GetDecals()
```

C++/CLI

```
System.Object^ GetDecals();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of [decals](#)

Visual Basic for Applications (VBA) Syntax

See [Component2::GetDecals](#).

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IComponent2::GetDecalsCount Method](#)
[IComponent2::!GetDecals Method](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetDecalsCount Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : GetDecalsCount Method (IComponent2)

Gets the number of decals applied to this component.

.NET Syntax

Visual Basic (Declaration)

```
Function GetDecalsCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IComponent2
Dim value As System.Integer

value = instance.GetDecalsCount()
```

C#

```
System.int GetDecalsCount()
```

C++/CLI

```
System.int GetDecalsCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of decals

Visual Basic for Applications (VBA) Syntax

See [Component2::GetDecalsCount](#).

Remarks

Call this method before calling [IComponent2::IGetDecals](#) to get the size of the array for that method.

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IComponent2::GetDecals Method](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetDrawingComponent Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : GetDrawingComponent Method (IComponent2)

Gets the drawing component for this component.

.NET Syntax

Visual Basic (Declaration)

```
Function GetDrawingComponent( _  
    ByVal ViewIn As View _  
) As DrawingComponent
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim ViewIn As View  
Dim value As DrawingComponent  
  
value = instance.GetDrawingComponent(ViewIn)
```

C#

```
DrawingComponent GetDrawingComponent(  
    View ViewIn  
)
```

C++/CLI

```
DrawingComponent^ GetDrawingComponent(  
& View^ ViewIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ViewIn

Pointer to [view](#) from which to get the drawing component

Return Value

Pointer to the [IDrawingComponent](#) object

Visual Basic for Applications (VBA) Syntax

See [Component2::GetDrawingComponent](#).

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetExcludeFromBOM2 Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : GetExcludeFromBOM2 Method (IComponent2)

Gets whether this component is excluded from the bills of materials (BOMs) in the specified configurations.

.NET Syntax

Visual Basic (Declaration)

```
Function GetExcludeFromBOM2( _  
    ByVal Config\_opt As System.Integer, _  
    ByVal Config\_names As System.Object _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim Config_opt As System.Integer  
Dim Config_names As System.Object  
Dim value As System.Object  
  
value = instance.GetExcludeFromBOM2(Config_opt, Config_names)
```

C#

```
System.object GetExcludeFromBOM2(  
    System.int Config\_opt,  
    System.object Config\_names  
)
```

C++/CLI

```
System.Object^ GetExcludeFromBOM2(  
& System.int Config\_opt,  
& System.Object^ Config\_names  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Config_opt

Configuration option as defined in [swInConfigurationOpts_e](#)

Config_names

Array of the names of configurations in whose BOMs this component is excluded or included; null or Nothing if Config_opt is set to swInConfigurationOpts_e.swAllConfiguration or swInConfigurationOpts_e.swThisConfiguration

Return Value

Array of values indicating whether this component is:

- excluded (true) from
- OR -

- included (false) in

the BOMs of the specified configurations.

Visual Basic for Applications (VBA) Syntax

See [Component2::GetExcludeFromBOM2](#).

Remarks

This method is valid only for [table-based bills of materials](#); it does not work for Microsoft Excel-based bills of materials.

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IAssemblyDoc::CompConfigProperties6 Method \(\)](#)
[IComponent2::SetExcludeFromBOM2 Method \(\)](#)

Availability

SOLIDWORKS 2019 FCS, Revision Number 27.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetHiddenUnloadedChildrenCount Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : GetHiddenUnloadedChildrenCount Method (IComponent2)

Gets the number of hidden children components of this component that were not loaded when an assembly was [opened selectively](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetHiddenUnloadedChildrenCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IComponent2
Dim value As System.Integer

value = instance.GetHiddenUnloadedChildrenCount()
```

C#

```
System.int GetHiddenUnloadedChildrenCount()
```

C++/CLI

```
System.int GetHiddenUnloadedChildrenCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of hidden children components of this component that were not loaded when an assembly was opened selectively

Visual Basic for Applications (VBA) Syntax

See [Component2::GetHiddenUnloadedChildrenCount](#).

Remarks

You can open an assembly with all components hidden except those that you select. The selected components are loaded lightweight. All other components become hidden and are not loaded into memory. This method returns the number of unloaded hidden children of this component. This method does not return the number of suppressed unloaded children components. Call [IComponent2::GetUnloadedComponentNames](#) for that type of related information.

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[ITreeControlItem Interface](#)
[IComponent2::HasUnloadedComponents Method](#)

Availability

GetID Method (IComponent2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : GetID Method (IComponent2)

Gets the component ID for this component.

.NET Syntax

Visual Basic (Declaration)

```
Function GetID() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IComponent2
Dim value As System.Integer

value = instance.GetID()
```

C#

```
System.int GetID()
```

C++/CLI

```
System.int GetID();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

ID for this component

Visual Basic for Applications (VBA) Syntax

See [Component2::GetID](#).

Example

[Get Component State \(C#\)](#)
[Get Component State \(VB.NET\)](#)
[Get Component State \(VBA\)](#)
[Get Component IDs \(C#\)](#)
[Get Component IDs \(VB.NET\)](#)
[Get Component IDs \(VBA\)](#)
[Get Top-level Components Using Component IDs \(C#\)](#)
[Get Top-level Components Using Component IDs \(VB.NET\)](#)
[Get Top-level Components Using Component IDs \(VBA\)](#)

Remarks

A component ID:

- is unique within the context of the assembly to which it belongs.
NOTE: A component ID might not be unique across subassemblies within the assembly. For example, a component in subassembly A might have a component ID of 1, and a component in subassembly B might also have a component ID of 1.
- is persistent across SOLIDWORKS Design sessions and never changes, even if you [change the name of the component](#).
- can be used to identify a specific component in an assembly. Use the component ID returned by this method with [IAssemblyDoc::GetComponentById](#) to get a top-level assembly component.
- cannot be assigned by applications or users.
- is not the same as a [persistent reference ID](#). You can get any component using its persistent reference ID.

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)

Availability

GetImportedPath Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : GetImportedPath Method (IComponent2)

Gets the full path name of the model imported for this component.

.NET Syntax

Visual Basic (Declaration)

```
Function GetImportedPath() As System.String
```

Visual Basic (Usage)



```
Dim instance As IComponent2
Dim value As System.String

value = instance.GetImportedPath()
```

C#

```
System.string GetImportedPath()
```

C++/CLI

```
System.String^ GetImportedPath();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Full path name of the model imported for this component; empty string if model is not imported

Visual Basic for Applications (VBA) Syntax

See [Component2::GetImportedPath](#).

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IComponent2::GetPathName Method \(\)](#)
[IModelDoc2::GetPathName Method \(\)](#)

Availability

SOLIDWORKS 2017 FCS, Revision Number 25.2

GetMaterialIdName Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : GetMaterialIdName Method (IComponent2)

Gets the material name for this component.

.NET Syntax

Visual Basic (Declaration)

```
Function GetMaterialIdName() As System.String
```

Visual Basic (Usage)



```
Dim instance As IComponent2
Dim value As System.String

value = instance.GetMaterialIdName()
```

C#

```
System.string GetMaterialIdName()
```

C++/CLI

```
System.String^ GetMaterialIdName();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Material name

Visual Basic for Applications (VBA) Syntax

See [Component2::GetMaterialIdName](#).

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IComponent2::SetMaterialIdName Method](#)
[IComponent2::GetMaterialUserName Method](#)
[IComponent2::SetMaterialUserName Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetMaterialPropertyValues2 Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : GetMaterialPropertyValues2 Method (IComponent2)

Gets the material properties for this component.

.NET Syntax

Visual Basic (Declaration)

```
Function GetMaterialPropertyValues2( _
    ByVal Config\_opt As System.Integer, _
    ByVal Config\_names As System.Object _
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IComponent2
Dim Config_opt As System.Integer
Dim Config_names As System.Object
Dim value As System.Object

value = instance.GetMaterialPropertyValues2(Config_opt, Config_names)
```

C#

```
System.object GetMaterialPropertyValues2(
    System.int Config\_opt,
    System.object Config\_names
)
```

C++/CLI

```
System.Object^ GetMaterialPropertyValues2(
& System.int Config\_opt,
& System.Object^ Config\_names
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Config_opt

Configuration option as defined in [swInConfigurationOpts_e](#)

Config_names

Array of configuration names for this component (see **Remarks**)

Return Value

The material properties values include the color (R,G,B values), reflectivity (ambient, diffuse, specular, shininess), transparency, and emission.

Visual Basic for Applications (VBA) Syntax

See [Component2::GetMaterialPropertyValues2](#).

Remarks

If you set *Config_opt* to *swThisConfiguration* or *swAllConfiguration*, then pass an empty Variant, Nothing, or any Variant because *Config_names* is ignored.

The format of *material_values* is an array of doubles:

[*R*, *G*, *B*, *Ambient*, *Diffuse*, *Specular*, *Shininess*, *Transparency*, *Emission*]

Valid values are from 0 to 1 for all variables. If *material_values* is all -1 values, then material property values were not assigned to the component. Therefore, the component will have the default properties in the user interface.

See Also

[IComponent2 Interface](#)

[IComponent2 Members](#)

[IComponent2::RemoveMaterialProperty2 Method](#)

[IComponent2::SetMaterialPropertyValues2 Method](#)

[IComponent2::GetMaterialPropertyValues2 Method](#)

[IComponent2::MaterialPropertyValues Property](#)

[IComponent2::GetModelMaterialPropertyValues Method](#)

[IComponent2::GetModelMaterialPropertyValues Method](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetMaterialUserName Method (IComponent2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : GetMaterialUserName Method (IComponent2)

Gets the user-visible name of the material for this component.

.NET Syntax

Visual Basic (Declaration)

```
Function GetMaterialUserName() As System.String
```

Visual Basic (Usage)

 Copy Code

```
Dim instance As IComponent2
```

```
Dim value As System.String
```

```
value = instance.GetMaterialUserName()
```

C#

```
System.string GetMaterialUserName()
```

C++/CLI

```
System.String^ GetMaterialUserName();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Material user name

Visual Basic for Applications (VBA) Syntax

See [Component2::GetMaterialUserName](#).

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IComponent2::GetMaterialIdName Method](#)
[IComponent2::SetMaterialIdName Method](#)
[IComponent2::SetMaterialUserName Method](#)

Availability

SolidWorks 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetMates Method (IComponent2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : GetMates Method (IComponent2)

Gets the mates for this component.

.NET Syntax

Visual Basic (Declaration)

```
Function GetMates() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim value As System.Object  
  
value = instance.GetMates()
```

C#

```
System.object GetMates()
```

C++/CLI

```
System.Object^ GetMates();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of mates (either [IMate2](#) objects or [IMateInPlace](#) objects)

Visual Basic for Applications (VBA) Syntax

See [Component2::GetMates](#).

Example

[Get Mates \(VB.NET\)](#)
[Get Mates \(VBA\)](#)
[Get Mates \(C#\)](#)

Remarks

This method eliminates iterating through the list of mates, which can adversely affect performance when the list of mates is long.

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetModelDoc Method (IComponent2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : GetModelDoc Method (IComponent2)

Obsolete. Superseded by [IComponent2::GetModelDoc2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetModelDoc() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim value As System.Object  
  
value = instance.GetModelDoc()
```

C#

```
System.object GetModelDoc()
```

C++/CLI

```
System.Object^ GetModelDoc();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

[Model document](#)

Visual Basic for Applications (VBA) Syntax

See [Component2::GetModelDoc](#).

Example

[Change Material Properties \(VBA\)](#)
[Get Display Dimensions, GTols, and Surface-finish Symbols \(VBA\)](#)
[Get BOM Balloon Properties \(C#\)](#)
[Get BOM Balloon Properties \(VB.NET\)](#)
[Get BOM Balloon Properties \(VBA\)](#)

Remarks

This method might return Nothing or null if:

- a component is suppressed or lightweight.
- the component ID is not loaded into memory by SOLIDWORKS.

For more information on lightweight components, see [Work With Lightweight Components](#).

When you use the IModelDoc2 object of a component, you do not have access to whatever uniqueness might exist for this instance of the assembly IComponent2. This occurs because the IModelDoc2 object goes back to the saved part file. By comparison, the IComponent2 object

gathers information at the assembly level. This allows IComponent2 objects to recognize assembly-level changes made to a component instance (for example, assembly-level features and material changes).

In addition, the IModelDoc2 object returned from this method represents the last-saved state. If the component part is currently open, then the IModelDoc2 object represents the state of the opened document. For example, if the component part is not open and it was last saved in the default configuration, then IComponent2::GetModelDoc returns a IModelDoc2 pointer representing that state. To get access to other configuration information (such as features and configuration properties), you must activate the part and show the desired configuration using [IModelDoc2::ShowConfiguration2](#).

If this component is the root component, then this method returns a NULL pointer. For more information, see [IConfiguration::GetRootComponent](#).

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IComponent2::IGetModelDoc Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetModelDoc2 Method (IComponent2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : GetModelDoc2 Method (IComponent2)

Gets the model document for this component.

.NET Syntax

Visual Basic (Declaration)

Function GetModelDoc2() As System.Object

Visual Basic (Usage)



```
Dim instance As IComponent2
Dim value As System.Object

value = instance.GetModelDoc2()
```

C#

```
System.Object GetModelDoc2()
```

C++/CLI

```
System.Object^ GetModelDoc2();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

[Model document](#)

Visual Basic for Applications (VBA) Syntax

See [Component2::GetModelDoc2](#).

Example

[Get Selected Objects and types in Assembly \(VBA\)](#)
[Insert MidSurface in Component \(C#\)](#)
[Insert MidSurface in Component \(VB.NET\)](#)
[Insert MidSurface in Component \(VBA\)](#)

Remarks

This method might return Nothing or null if:

- a component is suppressed or lightweight.
- the component ID is not loaded into memory by SOLIDWORKS.

For more information on lightweight components, see [Work With Lightweight Components](#).

When you use the IModelDoc2 object of a component, you do not have access to whatever uniqueness might exist for this instance of the assembly IComponent2. This occurs because the IModelDoc2 object goes back to the saved part file. By comparison, the IComponent2 object gathers information at the assembly level. This allows IComponent2 objects to recognize assembly-level changes made to a component instance (for example, assembly-level features and material changes).

In addition, the IModelDoc2 object returned from this method represents the last-saved state. If the component part is currently open, then the IModelDoc2 object represents the state of the opened document. For example, if the component part is not open and it was last saved in the default configuration, then IComponent2::GetModelDoc2 returns a IModelDoc2 pointer representing that state. To get access to other configuration information (such as features and configuration properties), you must activate the part and show the desired configuration using [IModelDoc2::ShowConfiguration2](#).

Unlike the previous version of this method ([IConfiguration::GetRootComponent](#)), this version of this method can handle the root component of an assembly.

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IComponent2::IsRoot Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetModelMaterialPropertyValues Method (IComponent2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IComponent2 Interface](#) : GetModelMaterialPropertyValues Method (IComponent2)

Gets the material properties of this lightweight component in the specified configuration.

.NET Syntax

Visual Basic (Declaration)

```
Function GetModelMaterialPropertyValues( _  
    ByVal ConfigName As System.String _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IComponent2  
Dim ConfigName As System.String  
Dim value As System.Object  
  
value = instance.GetModelMaterialPropertyValues(ConfigName)
```

C#

```
System.Object GetModelMaterialPropertyValues(  
    System.string ConfigName  
)
```

C++/CLI

```
System.Object^ GetModelMaterialPropertyValues(  
& System.String^ ConfigName  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ConfigName
Name of configuration

Return Value

An array of 9 doubles (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Component2::GetModelMaterialPropertyValues](#).

Example

[Get Model Material Property Values \(VBA\)](#)
[Get Model Material Property Values \(VB.NET\)](#)
[Get Model Material Property Values \(C#\)](#)

Remarks

The contents of the returned array:

[*R, G, B, Ambient, Diffuse, Specularity, Shininess, Transparency, Emission*]

Valid values are between 0.0 and 1.0, inclusive. Multiply the R, G, and B values by 255 to obtain the red, green, and blue component values. Multiply the other values in the array by 100 to obtain percentages. If all values in the array are -1, then material property values were not assigned to this component, and the component has the same default properties as the user interface.

See Also

[IComponent2 Interface](#)
[IComponent2 Members](#)
[IComponent2::GetMaterialPropertyValues2 Method](#)
[IComponent2::IGetModelMaterialPropertyValues Method](#)
[IComponent2::HasMaterialPropertyValues Method](#)

Availability

SOLIDWORKS 2011 FCS, Revision Number 19.0